



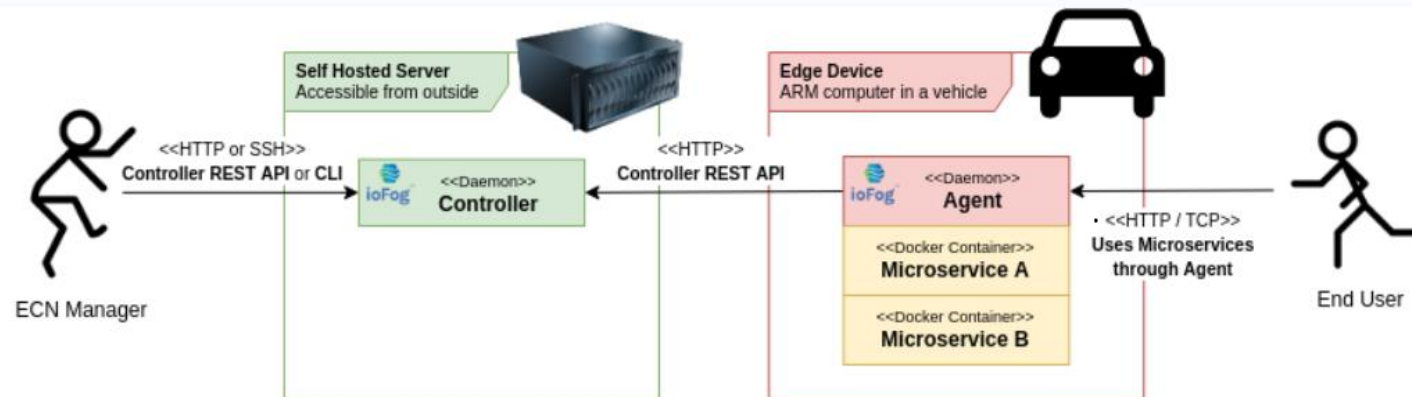
1. **ioFog - Edge Computing Framework** - open-source edge computing framework
2. **Novetta - Edge for Disaster response** – IoT solution is used to locate people, vehicles, and equipment during disaster.
3. **IoT-Driven Precision Agriculture** - deploys intelligence to fields so that farmers can gain visibility into crop yields and optimize their productivity and costs.
4. **Germany's DB Railway System** uses extensive IoT technologies to gather and communicate real-time passenger and cargo train information

ioFog - Edge Computing Framework



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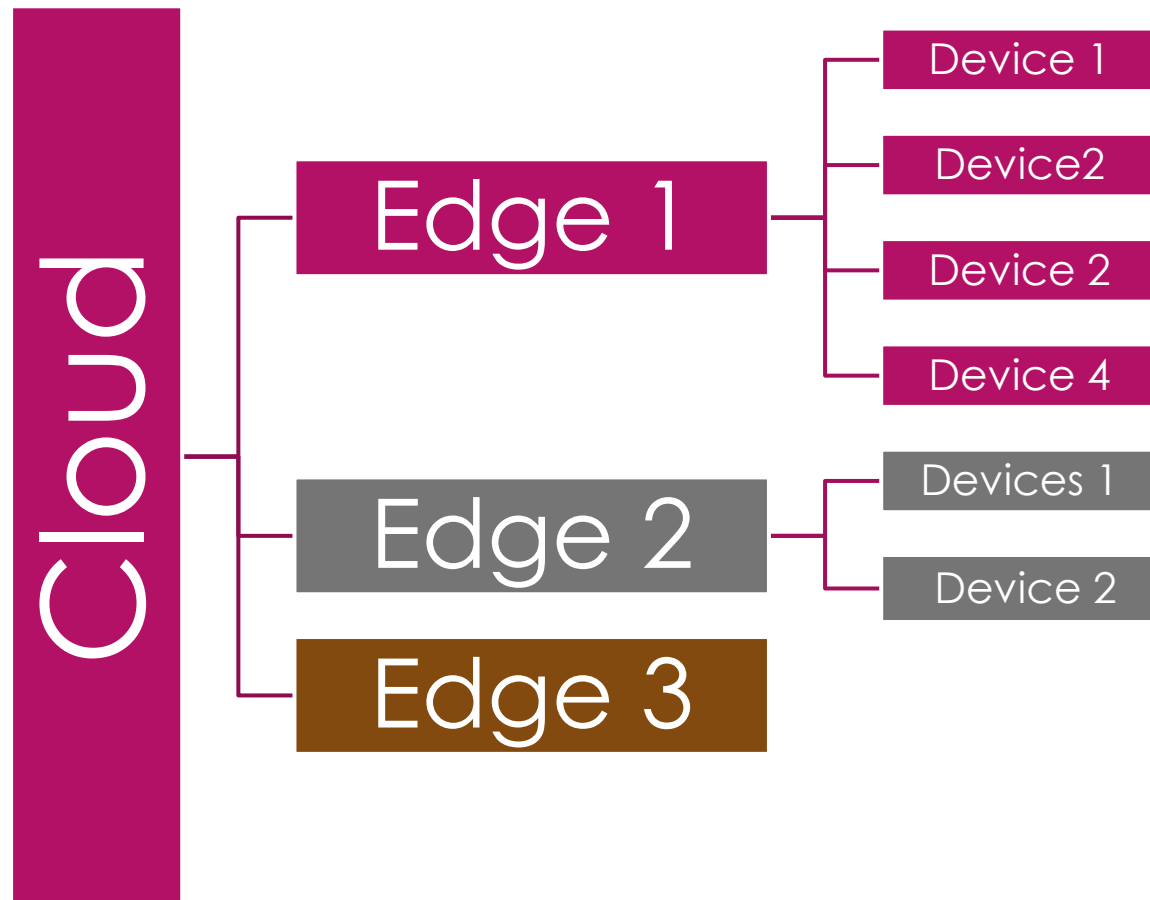
<https://iofog.org/docs/2/getting-started/architecture.html>



ioFog is an open-source edge computing framework. It supports a containerized microservices based runtime on edge servers, called Edge Compute Network (ECN). It uses different components like Agent, Controller, Router, Proxy, etc.

- **Controller** is the central orchestrator & heart of ECN managing all agents, microservices, routers users etc.
- **Agent** runs on the edge device (node) and is responsible for managing microservices, storage volumes, and other resources. It connects to a central controller which allows for remote orchestration of the overall setup.
- **Routers** enable microservices to communicate with each other. Each Controller & Agent would have it's own Router.
- **Proxy** is internal component on ioFog used to translate HTTP requests to AMQP Since the routers use AMQP

- Smart Electricity Grid - dynamic, responding to real time electricity consumption
 - Lowers production when it is not needed
 - would rely on real-time data of both production and consumption of electricity
 - household smart electricity meter



Novetta - Edge for Disaster response

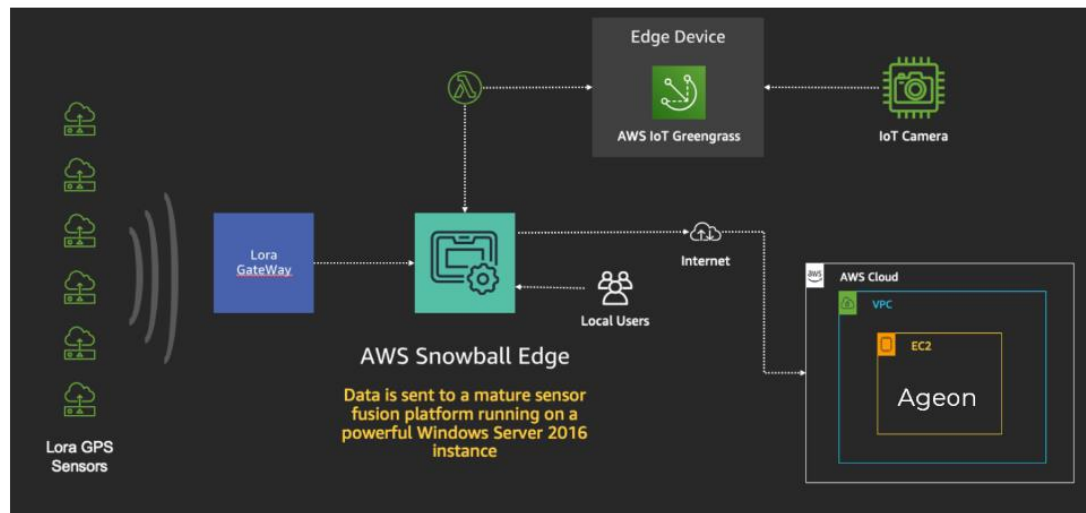


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During disaster response, maintaining high safety standards is paramount.

- Knowing the location of personnel, vehicles, and equipment is critical to maximizing the effectiveness of first responders and supporting safety efforts.
- Tracking the location of these resources requires participation from multiple organizations.
- Responding organizations often lack interoperability with local, state, and federal command centers, as well as with each other.
- Most available location tracking solutions depend on cellular or satellite networks, which can be unreliable, costly, and difficult to use during disaster.

<https://aws.amazon.com/blogs/storage/novetta-delivers-iot-and-machine-learning-to-the-edge-for-disaster-response/>



Novetta Solution

- Display tracking information using commodity GPS sensors dynamically assigned to people, vehicles, and equipment.
- LoRa protocol enables low-power, long-range encrypted communications between sensor and a central LoRa Gateway (Receiver & Packet Forwarder).
- Processing engine and the app server are running on Amazon EC2 instances launched on an AWS Snowball Edge Compute Optimized device. Snowball Edge provides a slice of your cloud environment in a rugged box that can function without being connected to the cloud.
- Sensor attribution is managed by a mobile application whose API is hosted on the Snowball Edge device in the field. The application tracks who and what is carrying the sensor, sensor health, reception, and battery status.

IoT-Driven Precision Agriculture



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<https://aws.amazon.com/blogs/iot/aws-iot-driven-precision-agriculture/>

- Precision agriculture is deploying intelligence to fields so that farmers can gain visibility into crop yields and cost factors as they grow crops.
- Intelligence is collected by smart devices or IoT things equipped with sensors that continuously act on the data and transmit it to data processing locations.
- Farmers are using newly developed techniques to
 - manage crops of their fields to assess soil composition, control fertilizer and pesticide types, and obtain accurate yield estimates.
 - understanding of the main causes for changes in crop yields and production cost drivers, including the following:
 - Fertilizers and their application timing and methods.
 - Irrigation and precipitation patterns.
 - Air temperature and moisture levels.
 - Soil composition and treatment.
 - Crop care processes.
 - Types of seeds used.

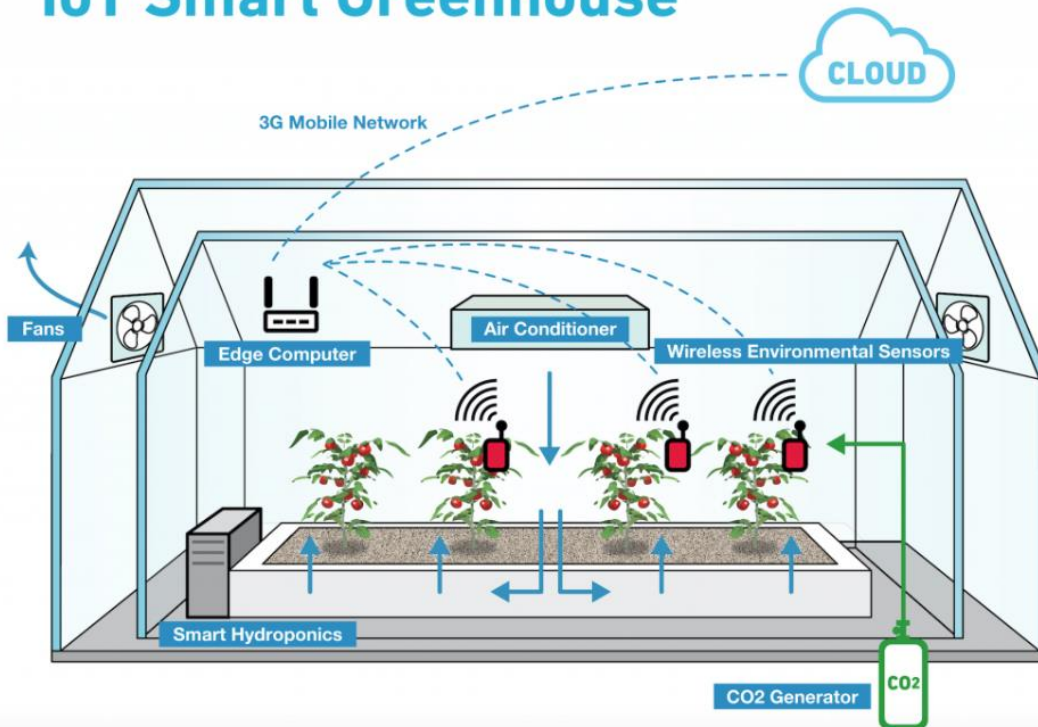
IoT-Driven Precision Agriculture



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IoT Smart Greenhouse



Yanmar's R&D group would like to make their greenhouse operations smarter by automatically detecting and recognizing the main growth stages of vegetables in order to adjust nutrition, care, and environment to maximize yields.

Yanmar has defined several growth stages for tomato plants.

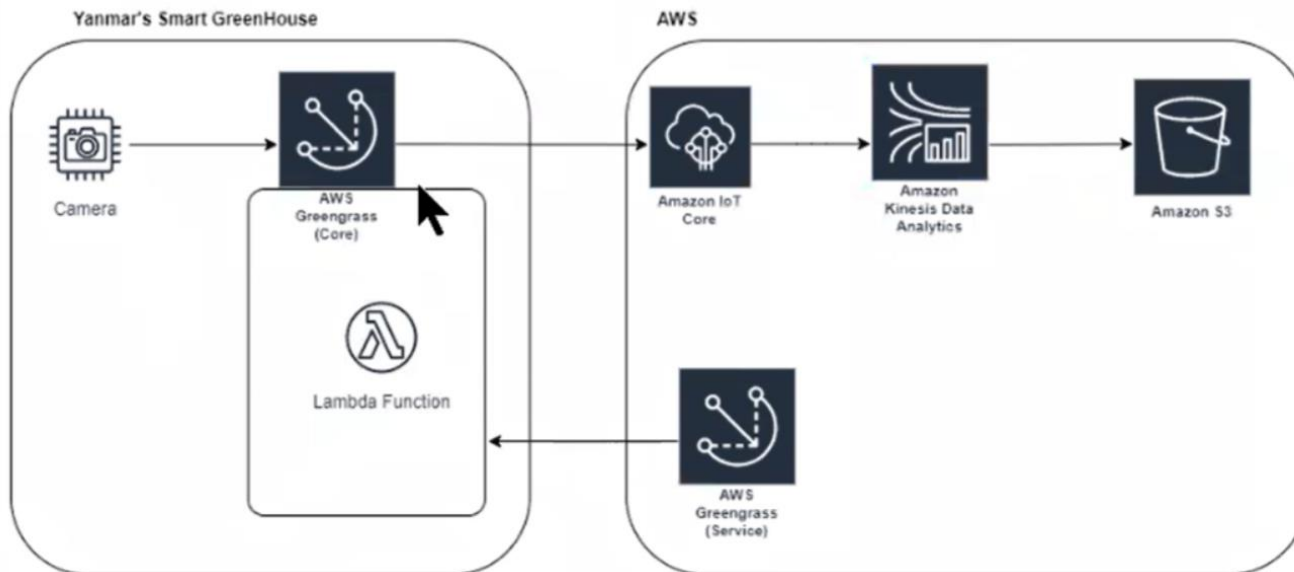
- To monitor these stages, they plan to install up to 12 cameras in each greenhouse to take pictures of the plants on a regular basis.
- Yanmar plans to use machine learning (ML) algorithms deployed on the camera ecosystem to recognize (by height, number of leaves, number of flowers, and fruits) plant growth stages and perform attribution against plant care factors, such as hydroponics application patterns and frequency, in addition to moisture and temperature.

IoT-Driven Precision Agriculture



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Deploy AWS Greengrass ML Inference using Lambda functions, on the greenhouse camera ecosystem because it addresses the following key challenges.

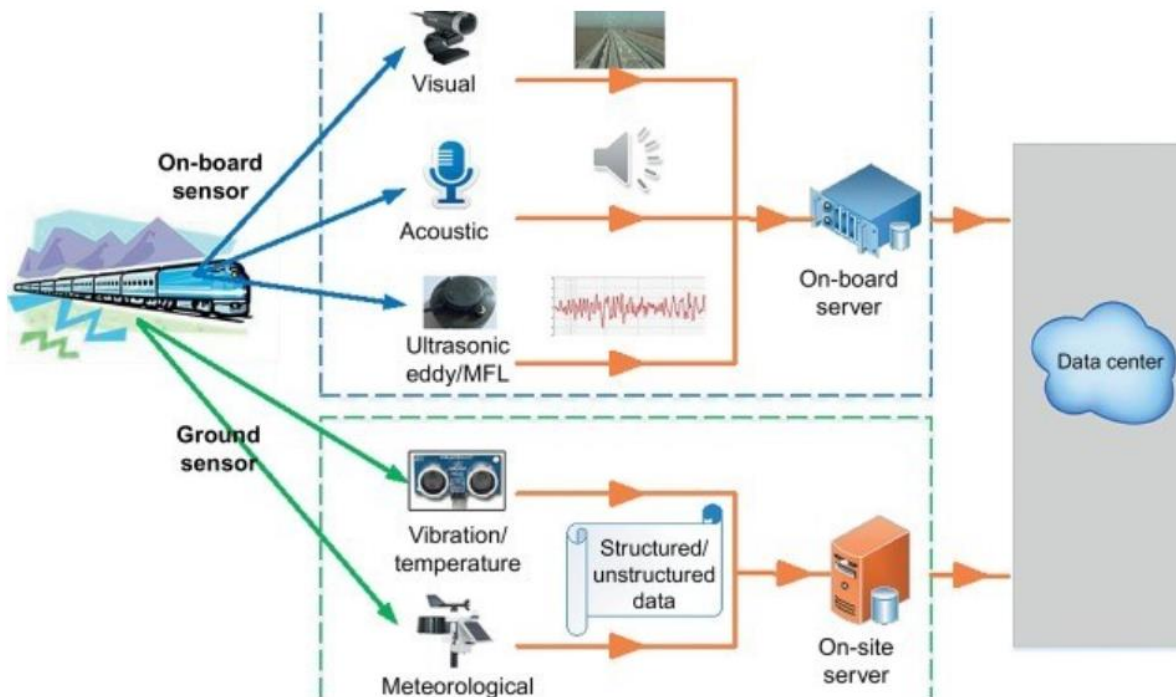
1. No need need to send the pictures collected from the cameras to AWS IoT Core over a 3G network, thereby avoiding 3G usage fees.
2. Greengrass accesses trained and deployed deep learning ML models to process greenhouse plant pictures and store the data in a simple format for transmission to AWS IoT Core.
3. Use Lambda functions to perform logic processing using local ML models to detect environment anomalies (moisture, temperature, and humidity) and trigger both alerts and possibly initiate corrective actions.

Germany's DB Railway System



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<https://iot.eclipse.org/community/resources/case-studies/pdf/Eclipse%20IoT%20Success%20Story%20-%20DB.pdf>



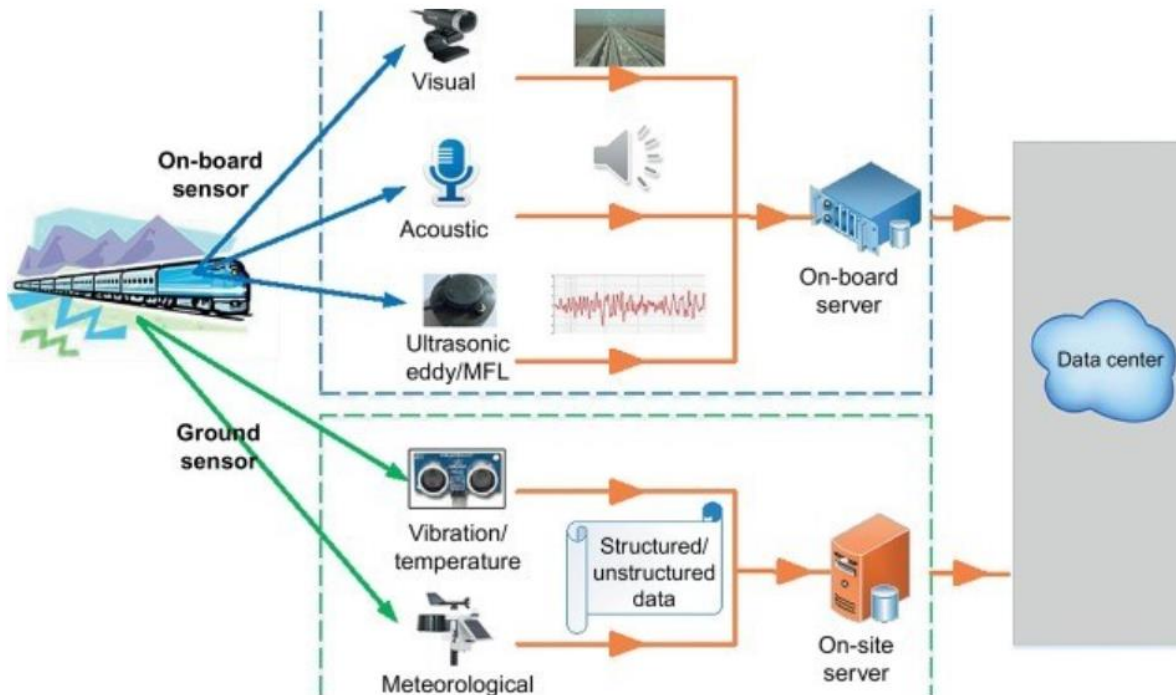
- Germany's DB Railway System uses extensive IoT technologies to communicate real-time passenger and cargo train locations, provide information on consumer displays, and ingest status of escalators and elevators for monitoring and predictive maintenance.
- Train location tracking –**
 - Trains transmit location data every 10 seconds to a central IBM MessageSight server, over MQTT.
 - In addition, they periodically send diagnostic data, track changes, and delay information.
 - Extensively uses MQTT for bidirectional communication. It uses Eclipse Paho and Eclipse Mosquitto as client libraries for various interactions with the central MQTT broker
 - All of this is correlated with running times to provide immediate delay forecasting to various consumers.
 - The diagnostic data also helps in predictive maintenance and track co-ordination.

Germany's DB Railway System



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<https://aws.amazon.com/blogs/iot/aws-iot-driven-precision-agriculture/>



Escalators and Elevators

- Around 3000 edge devices connected with various sensors send 10 messages per second to the central server. This includes running state information, door open/close status, power consumption levels, and diagnostic data. This helps proactive and predictive long-term provides an API for app development to guide users.

Data processing and display

- Central server processes & correlates all this data, resulting in multiple insights including delay forecasting, platform change information, imminent arrivals, and long-term maintenance analysis

The consumer specific data is relayed over multiple channels

- Transmitted back to the trains via MQTT, to convey to train passengers over internal electronic displays.
- Transmitted to low powered consumer displays on the train stations, which act as MQTT subscribers.
- Mobile apps and websites created using the APIs providing all this important data to potential travelers, wherever they may be.